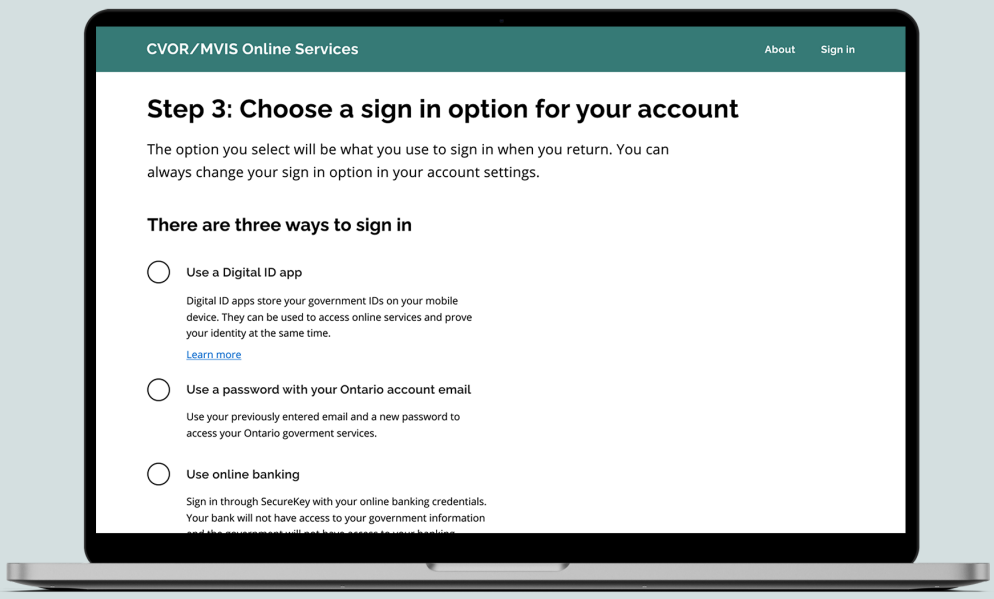
Public Secure SSO

PRODUCT DESIGN — 2020

I worked as a Product Designer with the Ontario Digital Service Prototyping Lab to help lay the groundwork for a new single sign-on (SSO) for the province's digital services. We worked alongside 13 different program areas and the Public Secure team before handing our work off to a vendor.



Scope & context.

As put by an end-user we spoke with, the current one key single sign-in portal acts as a 'fake front door', in that most of the program areas using ONe-key require further verification upon the completion of signing in. Some of the larger issues we tackled were:

- Old ONe-key accounts needed be effectively transferred to newer Public Secure accounts

- Developing Public Secure sign in and sign up flows that were consistent and provided the user with a much more seamless experience when it comes to logging on with Public Secure through a program areas application – This involved creating a set of standardized flows for all three different levels of authentication required by program areas (low, medium, high).

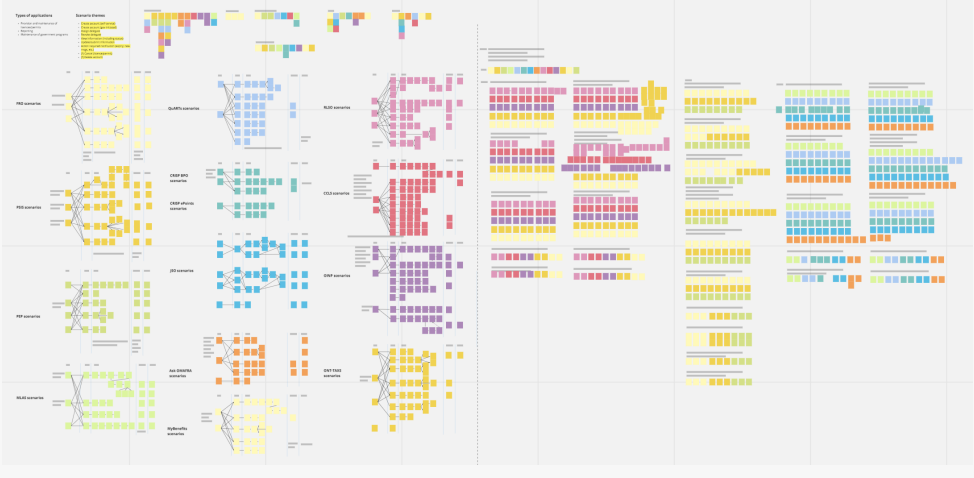
- Wording and instruction surrounding naming of the signing (going from ONe-key, to Public Secure, to Ontario Account) as well as around the description of the different sign-in options (making sure the options of signing in with a password versus with online banking versus with a Digital ID app (which is very future-state and not a current option) was clear for users).

PROBLEM STATEMENT

We observed that ONe-key and Public Secure Pilot confuse our public users while not meeting the expectations of our internal stakeholders, which causes a lower adaption of Ontario's single sign-on solution from digital services, in addition to higher call centre volumes from users. Our goal was to improve Ontario's single sign-on solution so that more program areas and end-users would adopt Public Secure, and so that call centre volumes would subsequently be reduced.

STAKEHOLDER INTERVIEWS

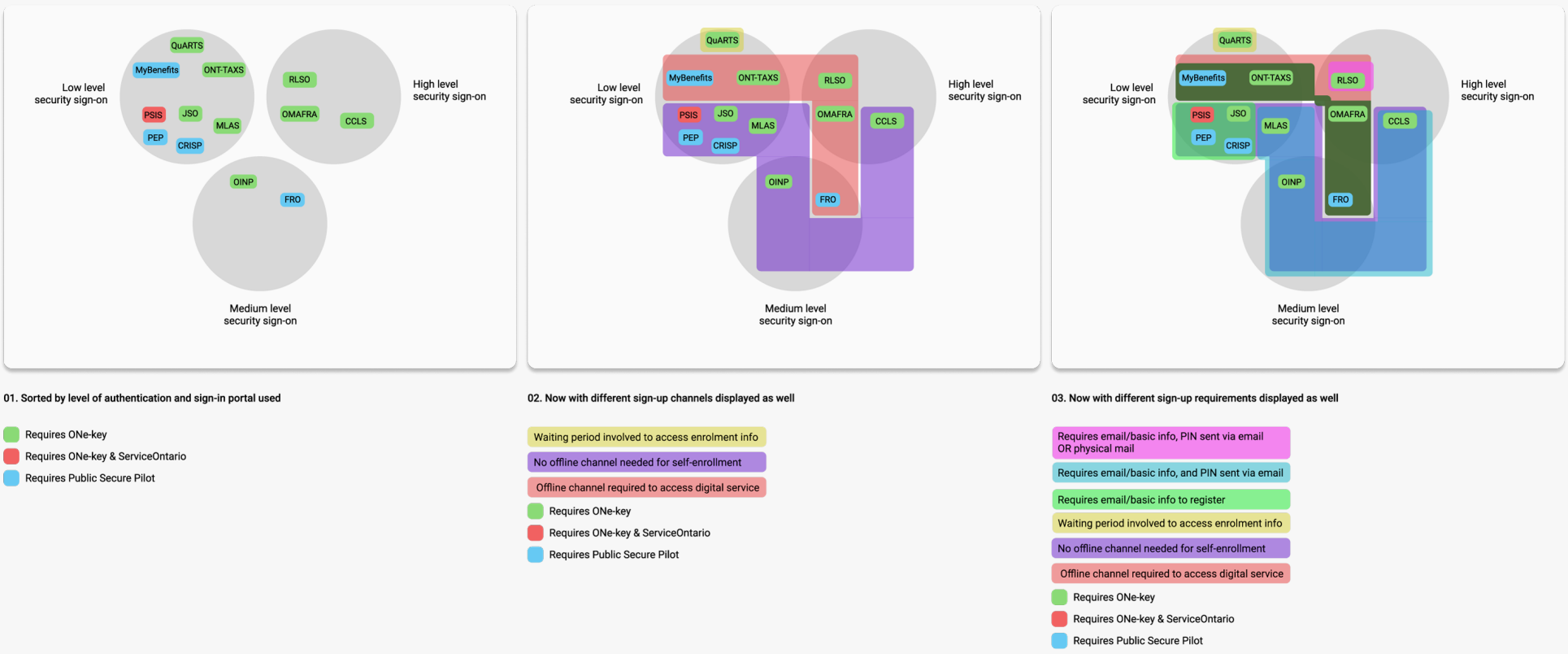
We talked to 13 program areas, composed of eight applications currently using ONe-key, four on the Public Secure Pilot, and one on a portal that uses both ONe-key and Service Ontario. We identified some commonly mentioned pain points that existed within ONe-key and Public Secure Pilot.



- **Password recovery:** Poor system in place for password recovery increases the number of lost accounts.
- **Content and wording:** Ambiguous wording/naming surrounding sign-in options confuses users.
- **User flows** Inconsistent onboarding and sign-on flows make for a disjointed experience.

AUTHENTICATION LEVELS

There are three unique levels of authentication: low (username and password), medium (username, password and an additional piece of ID), and high (username, password, two-factor authentication). Depending on the program area being accessed, different pieces of information were required within and throughout these three levels of authentication to sign-up or sign-in, making the concept of a single sign-on tricky.

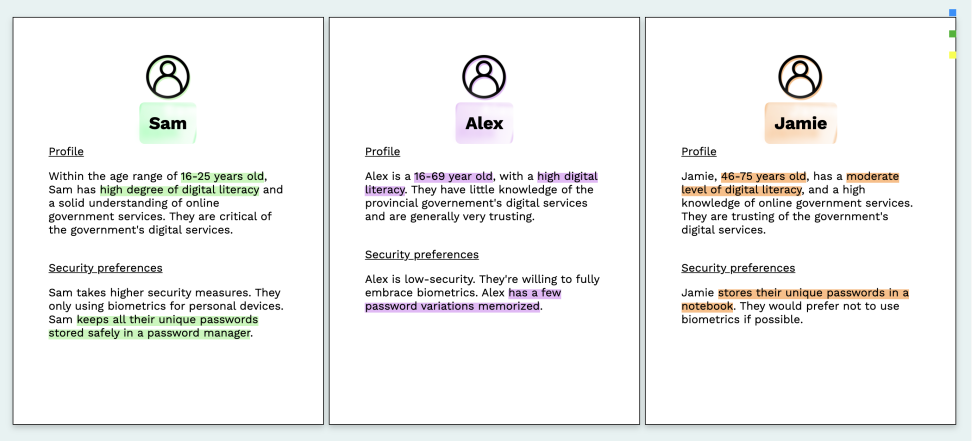


Authentication levels, channels, and required materials visualized – you can see how it gets pretty messy.

USER INTERVIEWS & PERSONAS

As this was a project occurring in the midst of the COVID-19 pandemic, recruiting users to interview proved to be a challenge. After the synthesis of our 12 user interviews, we grouped common traits and came up with three user personas. As this is a single sign-on that has the potential to be used by effectively any member of the Ontario public, we

chose not to create personas within a specific niche, instead opting for a more all-encompassing set of personas to help steer our work.



Sam

Profile

Within the age range of 16-25 years old, Sam has high degree of digital literacy and a solid understanding of online government services. They are critical of the government's digital services.

Security preferences

Sam takes higher security measures. They only using biometrics for personal devices. Sam keeps all their unique passwords stored safely in a password manager.

Alex

Profile

Alex is a 16-69 year old, with a high digital literacy. They have little knowledge of the provincial governement's digital services and are generally very trusting.

Security preferences

Alex is low-security. They're willing to fully embrace biometrics. Alex has a few password variations memorized.

Jamie

Profile

Jamie, 46-75 years old, has a moderate level of digital literacy, and a high knowledge of online government services. They are trusting of the government's digital services.

Security preferences

Jamie stores their unique passwords in a notebook. They would prefer not to use biometrics if possible.

Personas based off user iterviews

User experience design.

After synthesizing stakeholder and user interviews notes we drafted up flows we found were particularily common and/or problematic.

1. Net new (account creation)
2. Needs to sign-in
3. Can't remember password
4. Has a ONe-key account (migrating account)
5. Single sign-on account management

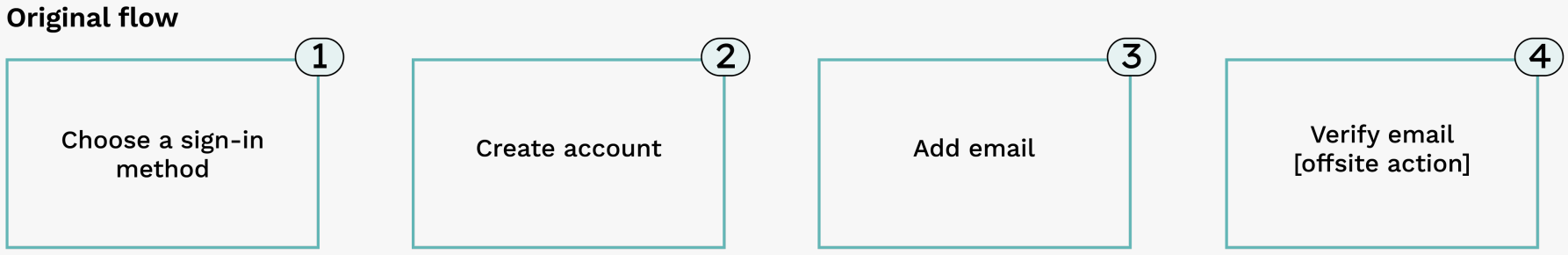
We ultimately drafted numerous flows that encompassed everything from informing Public Secure users of the portal changes to incorrect sign-on attempts, but those five flows remained the most important throughout our testing sessions.

SIGN-UP MADE SIMPLER

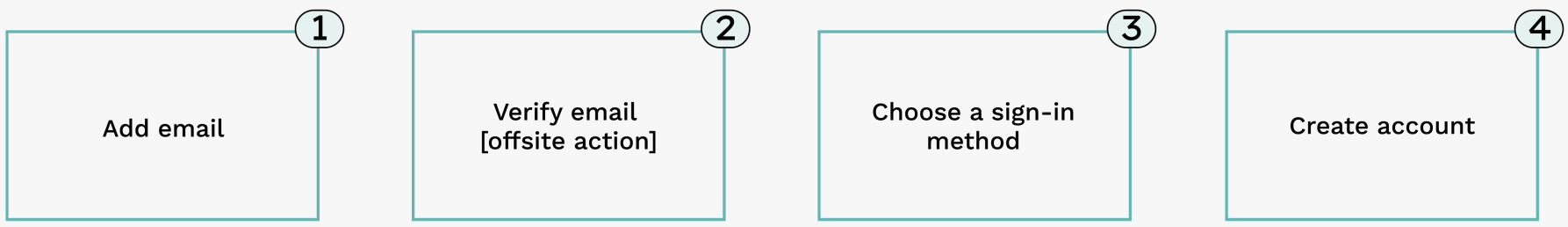
Shortly after we began user testing with our preliminary onboarding flows, we discovered that many users confused the different sign-in options as different types of accounts altogether. Since emails were mandatory regardless of the chosen sign-in method, we decided to use the email to initiate the account creation proccess. On top of simplifying the flows, this changed the way many users perceived the sign-in options: now instead of seeing them as different account

types, they more appropriately recognized as different means of signing in to the same account.

Original flow



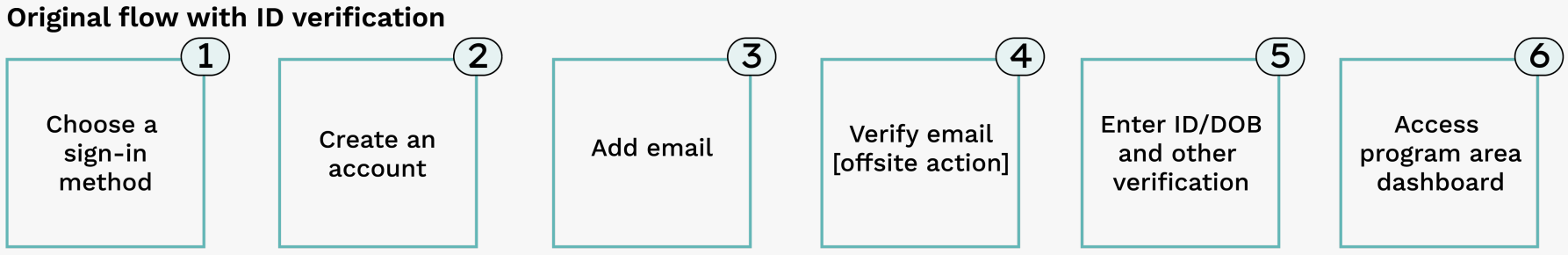
New flow



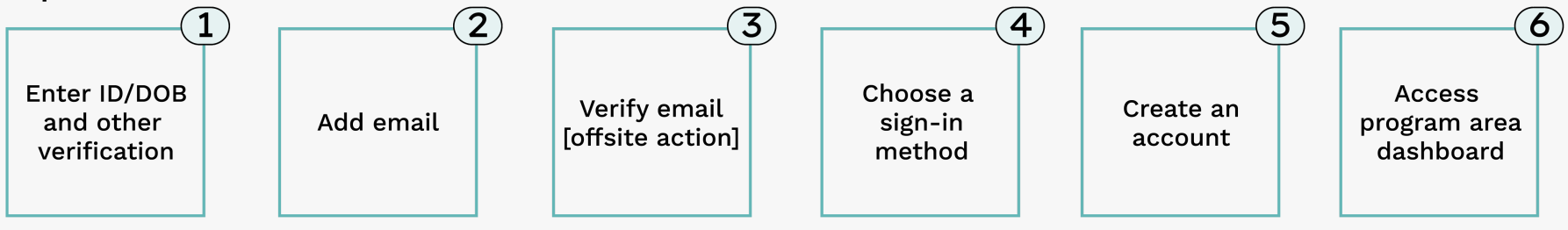
MANAGING ADDITIONAL ID'S

During our testing sessions many users confused unique ID's (ie. MyBenefits' MemberID) with something needed to create/sign-in to an Ontario account, while some outright forgot that it was even a part of the registration process to begin with by the time the screen for it arrived. We changed it so that the user is prompted to input their unique ID before creating an account.

Original flow with ID verification



Improved flow with ID verification

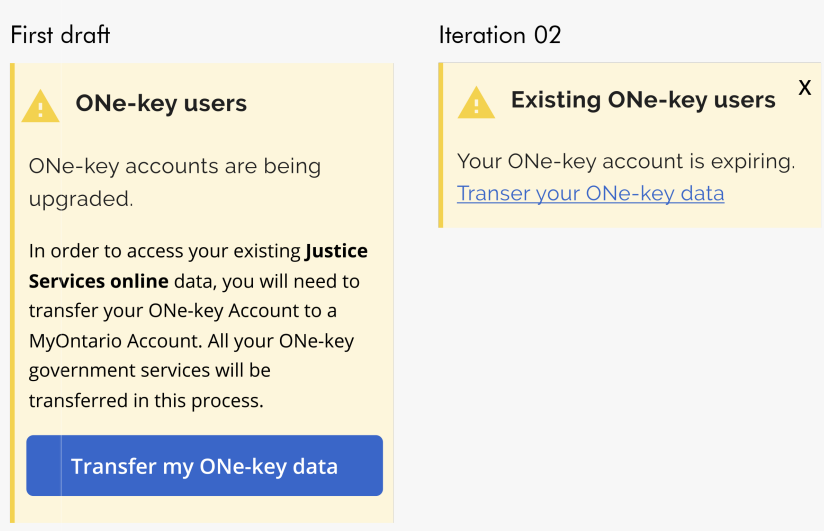


UI design.

As we ran through 11 user testing sessions we ended up with five iterations of our prototypes. In the ensuing section I'll detail a few more, interface related, changes we made to our screens.

ITERATIONS 00 - 02: ONE-KEY ALERT

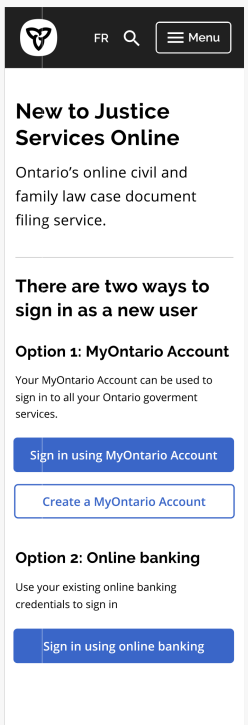
On our initial mockups we had a yellow banner, warning ONe-key users that their account needed to be transferred. Many non-ONe-key users we tested with were unable to determine whether this warning was relevant to them or not. We changed the warning box to be less verbose in future iterations, reducing the potential for confusion from those unfarmiliar with ONe-key, and increasing the likeliness of a user actually reading it, in case it was pertinent to them.



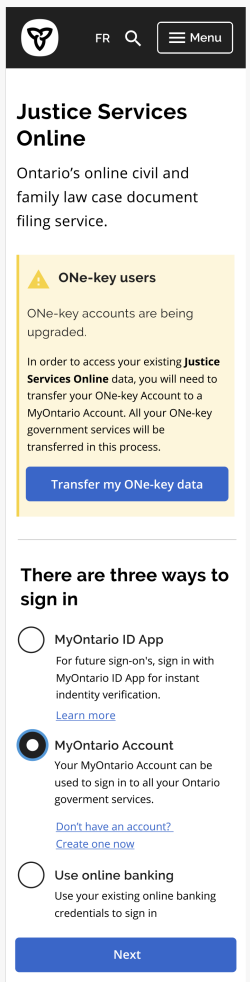
ITERATIONS 00 - 02: IMPROVED BUTTONS

Another significant change made was the buttons on the landing page. Based on stakeholder feedback and then the subsequent first user tests, we were able to confirm that having primary and secondary buttons inherently created a hierarchy within the options, which are all of equal importance. We changed these buttons to radio buttons and soon after noticed that this users were more inclined to thoroughly read through all their options.

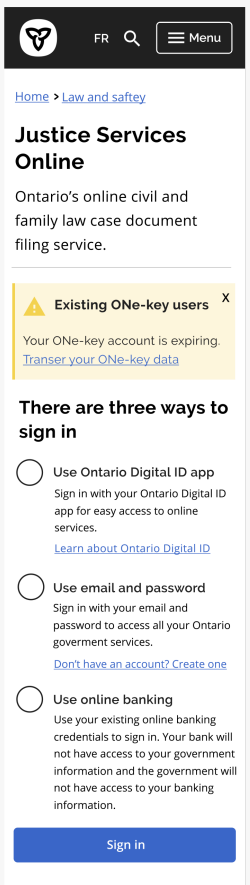
Preliminary



Iteration 01



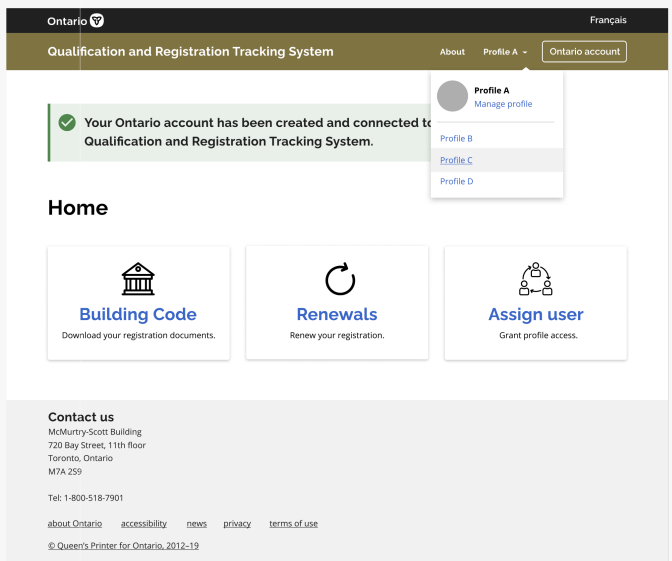
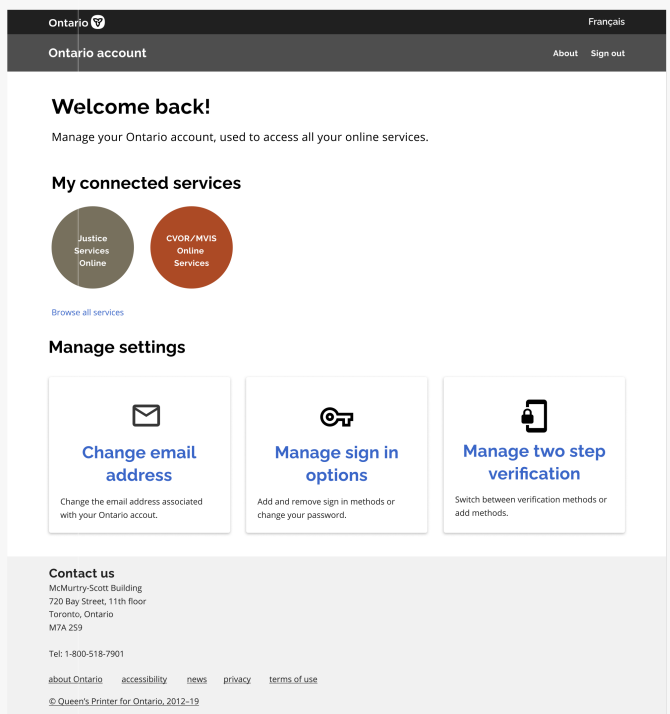
Iteration 02



ITERATIONS 03 - 05: ACCOUNT MANAGEMENT

With a single sign on for government services, users need the ability to manage profiles within a service as well as their overall account. For signed in users we needed a dropdown menu for the user to manage account settings/sign out as well, and for navigating different profiles within a service we needed a second dropdown menu.

Profile selections



Central account management